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**COURSE NAME: SOFTWARE
ENGINEERING**

ASSESSMENT TOOL - 02

Project Title

**Agile Project Planning – AI-Powered Personalized Nutrition and Health
Recommendation Platform**

1. Project Vision and Stakeholders

Project Vision

To develop an AI-powered web/mobile platform that provides personalized nutrition plans, health recommendations, and continuous health monitoring based on individual health data, dietary preferences, lifestyle, and wearable device information.

Project Objectives

- Collect personal health information securely.
- Analyze nutritional requirements using AI.
- Generate personalized meal plans.
- Monitor daily dietary intake.
- Integrate wearable health devices.
- Provide AI-based health recommendations.
- Track nutrition and fitness progress.
- Encourage healthy lifestyle habits.

Key Stakeholders

Stakeholder	Role
End Users	Receive personalized nutrition and health recommendations
Nutritionists	Validate meal plans and provide expert guidance
Doctors	Monitor patient health and provide recommendations
System Administrator	Manage users, security, and platform maintenance
AI/ML Development Team	Build AI recommendation engine

Stakeholder	Role
Mobile/Web Developers	Develop application
Database Administrator	Maintain health records securely
Project Manager	Manage project execution
Fitness Device Providers	Supply wearable health data

2. Epics and User Stories

Epic 1: User Management

User Story 1

As a new user,
I want to register my account,
so that I can access personalized nutrition services.

User Story 2

As a registered user,
I want to log in securely,
so that my health information remains protected.

Epic 2: Health Profile

User Story 3

As a user,
I want to enter my age, height, weight, medical history and dietary preferences,
so that the AI can generate personalized recommendations.

Epic 3: AI Nutrition Recommendation

User Story 4

As a user,
I want AI-generated meal plans,
so that I can follow a healthy diet.

User Story 5

As a user,
I want daily calorie recommendations,
so that I maintain a balanced diet.

Epic 4: Food Tracking

User Story 6

As a user,
I want to record my meals,
so that I can monitor my nutrition intake.

Epic 5: Wearable Integration

User Story 7

As a user,
I want to connect my smartwatch,
so that my health data is updated automatically.

Epic 6: Progress Monitoring

User Story 8

As a user,
I want to view my nutrition progress,
so that I stay motivated to improve my health.

3. Acceptance Criteria (Given-When-Then)

User Story 1 – Registration

Given the user is on the registration page
When valid details are entered
Then the account should be created successfully.

User Story 2 – Login

Given the user has registered
When correct username and password are entered
Then the dashboard should open.

User Story 3 – Health Profile

Given the user is logged in
When health information is submitted
Then the system should save the profile successfully.

User Story 4 – AI Meal Plan

Given the user's profile is complete
When the user requests a recommendation

Then AI should generate a personalized meal plan.

User Story 5 – Calorie Recommendation

Given the user's BMI and activity level

When AI performs analysis

Then recommended daily calories should be displayed.

User Story 6 – Meal Tracking

Given the user has eaten food

When the meal is logged

Then nutrition values should update automatically.

User Story 7 – Wearable Integration

Given a wearable device is connected

When health data syncs

Then activity information should be updated.

User Story 8 – Progress Tracking

Given the user has nutrition history

When the dashboard opens

Then charts and progress reports should be displayed.

4. Product Backlog (Prioritized)

Priority	User Story	Business Value
High	User Registration	Very High
High	Secure Login	Very High
High	Health Profile	Very High
High	AI Meal Recommendation	Very High
Medium	Meal Tracking	High
Medium	Calorie Recommendation	High
Low	Wearable Integration	Medium
Low	Progress Dashboard	Medium

5. Sprint Backlog (Sprint 1)

Sprint Duration

2 Weeks

User Story	Development Tasks
Registration	Design UI, Validate Form, Store User Data
Login	Authentication, Password Encryption
Health Profile	Design Form, Database Tables, Validation
AI Meal Recommendation	Develop Recommendation Algorithm, Display Meal Plan

6. Story Point Estimation (Fibonacci)

User Story	Story Points	Justification
Registration	3	Simple CRUD operations
Login	3	Authentication and security
Health Profile	5	Multiple health fields and validation
AI Meal Recommendation	8	AI logic and recommendation engine
Meal Tracking	5	Food database integration
Calorie Recommendation	5	Calculation algorithms
Wearable Integration	13	External API integration
Progress Dashboard	8	Charts, reports, analytics

Why Fibonacci?

- 1 = Very simple
- 2–3 = Small feature
- 5 = Moderate complexity
- 8 = Complex
- 13 = Highly complex with external dependencies

7. Sprint Goal and Expected Deliverables

Sprint Goal

Develop the core user management and AI recommendation features so users can register, log in, create a health profile, and receive personalized meal recommendations.

Expected Deliverables

- ☒ User Registration Module
- ☒ Secure Login System
- ☒ Health Profile Management
- ☒ AI-Based Meal Recommendation Engine
- ☒ User Dashboard
- ☒ Backend Database Integration
- ☒ Basic Testing and Bug Fixes

This Sprint 1 establishes the platform's core functionality and provides a working prototype capable of delivering personalized nutrition recommendations.